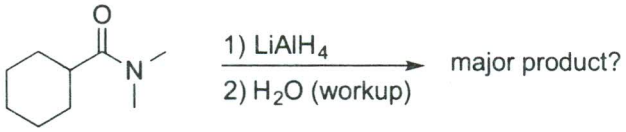
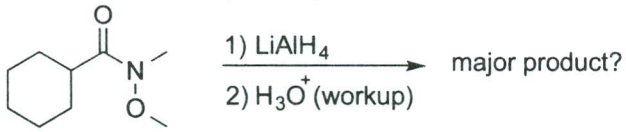
Organic Cume 04-24-2015

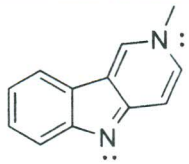
1. a) Give the major organic product of the following reaction. Write an electron-pushing mechanism to account for its formation.



b) Weinreb amide shown below is designed to give a different product. What is it? Write an electron-pushing mechanism to account for its formation.

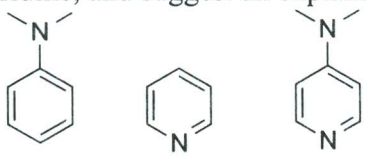


2. The compound shown is a somewhat stronger base than ammonia. Which nitrogen do you think is protonated when it is treated with an acid? Write a structural formula for the species that results.



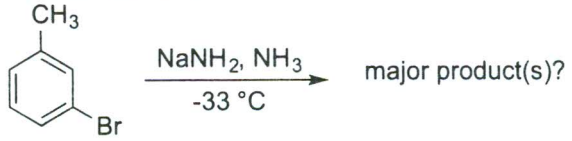
5-methyl-γ-carboline
p*K*a of conjugate acid = 10.5

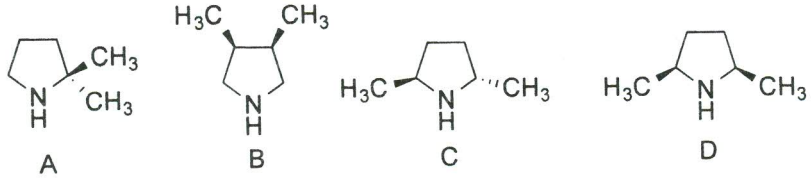
3. *N,N*-Dimethylaniline and pyridine are similar in basicity, whereas 4-(*N,N*-dimethylamino)pyridine is considerably more basic than either. Apply resonance principles to identify the more basic of the two nitrogen atoms of 4-(*N,N*-dimethylamino)pyridine, and suggest an explanation for its enhanced basicity



p*K*a of conjugate acid	5.1	5.3	9.7

4. Give the product(s) of the following reaction. Write an electron-pushing mechanism to account for the product formation.



5. (+)-2-Allylcyclohexanone has been prepared in 82% enantiomeric excess by alkylation of the optically active enamine prepared from cyclohexanone and an enantiomerically

pure pyrrolidine derivative. Of the following, which one is the best pyrrolidine derivative to use in this enantioselective synthesis? Give the reason for your choice briefly.